

In-context perturbation world model

Current stability setup: STACK state embeddings + Borzoi action embeddings -> set transformer -> perturbed state

Inputs and task construction

AnnData .obsm states

X_stack, cells x 1600
cached STACK embeddings
upstream model frozen/offline

AnnData .obsm actions

X_pert_borzoi_v0
gene/action vectors, dim=1536
lookup table is frozen

In-context sampler

support: M=16 triplets
query: one held-out perturbation
bucket: cell_line + batch

Batch tensors

support_obs, support_next, support_act
query_obs, query_act
query_next target

Encoders

State head: ForeignBackboneHead

mean-pool stack K=4
Linear 1600 -> 256
trainable params: 409,856

Latent normalization

none | LayerNorm | I2
applied before dynamics/loss

State tokens

support_s, support_sp, query_s

GeneEmbeddingAction

frozen embedding lookup
ActionAdapter Linear 1536 -> 256
masked mean over n_pert=2

Action tokens

support_a, query_a

InContextSetDynamics

Triplet fusion

support: concat(s, a, s')
query: concat(query_s, query_a, MASK)
Linear + GELU + LayerNorm

Bidirectional set transformer

6 TransformerBlock layers, 8 heads
bidirectional; no positional embeddings
role embedding marks query token

Query token head

LayerNorm -> Linear(256, 256)
trainable params: 5,003,008

Support selection

random or action_similarity
same-bucket constraints retained

Prediction and loss

Prediction mode

absolute: raw = s_hat
residual_delta: raw = delta_hat
then s_hat = query_s + delta_hat

ExpressionDecoder

MLP 256 -> 512 -> 1024 -> 1600
outputs x_hat

Training losses

latent objective: MSE
residual target: s_target - query_s
decoder_mse on query_next_expression
InfoNCE on deltas
action-counterfactual CE

Post-fit evaluation

CPU cell_eval profile=full
baselines: identity, mean, additive, linear